

# Sharp partial-pivoting growth for unequal bandwidths

*Reconstructed candidate proof draft. Independent mathematical review and source/priority verification are required. The IE identifiers are provisional labels recovered from earlier notes, not verified current repository identifiers.*

## Abstract

For lower bandwidth  $p \geq 1$  and upper bandwidth  $q \geq 0$ , the proposed optimal dimension-independent GEPP growth factor is  $h_{p+q}$ , where  $h_t = 0$  for  $t \leq 0$  and  $h_t = 1 + \sum_{r=1}^p h_{t-r}$  for  $t \geq 1$ . A sorted-front argument gives an upper bound, and an explicit nonsingular real band matrix attains it. The proof covers fixed column order, real and complex matrices, and arbitrary permitted pivot ties.

## 1 Statement

Let  $A \in \mathbb{R}^{n \times n}$  or  $\mathbb{C}^{n \times n}$  be nonsingular, with  $a_{ij} = 0$  whenever  $i > j + p$  or  $j > i + q$ . Normalize  $\max |a_{ij}| = 1$ . Growth includes all active Schur-complement entries, not merely  $U$ . Define  $G(p, q)$  as the supremum over permitted orders and pivot paths.

**Theorem 1.**  $G(0, q) = 1$ . For  $p \geq 1$ ,

$$h_t = 0 \quad (t \leq 0), \quad h_t = 1 + \sum_{r=1}^p h_{t-r} \quad (t \geq 1), \quad \boxed{G(p, q) = h_{p+q}}.$$

A real matrix of order  $2p + q + 1$  attains the bound.

The recurrence gives

$$\begin{aligned} G(p, 0) &= 2^{p-1}, \\ G(p, q) &= 2^{p+q-1} - (q-1)2^{q-2} \quad (1 \leq q \leq p+1), \\ G(1, q) &= q+1, \quad G(2, q) = F_{q+4} - 1, \end{aligned}$$

where  $F_0 = 0, F_1 = 1$ . In particular the equal-bandwidth specialization is  $2^{2p-1} - (p-1)2^{p-2}$ . These identities follow by induction.

## 2 Sorted-front lemma

Suppose a front has  $p+1$  target-column magnitudes  $y_1 \geq \dots \geq y_{p+1} \geq 0$ . Remove the row with target magnitude  $y_a$  and update the others with multipliers of modulus at most one. The sorted survivors are bounded componentwise by

$$(y_1 + y_2, y_1 + y_3, \dots, y_1 + y_{p+1}).$$

Indeed, the triangle inequality bounds each survivor by  $y_a + y_i$ . For  $i < a$ ,  $y_a + y_i \leq y_1 + y_{i+1}$ , since  $y_i - y_{i+1} \leq y_1 - y_a$ . For  $i \geq a$ , the ordered bound is  $y_a + y_{i+1} \leq y_1 + y_{i+1}$ . Sorting and appending a fresh value are monotone. Thus always eliminating the largest target-column

magnitude gives a componentwise upper envelope, even though GEPP actually chooses by another column.

Start with  $p$  zeros, append a fresh 1, and perform this envelope update. After one update all survivors equal 1. Subsequently the ordered state evolves by

$$(x_1, \dots, x_p) \mapsto (x_1 + x_2, \dots, x_1 + x_p, x_1 + 1).$$

After  $t \geq 1$  updates its  $i$ th entry is

$$x_i^{(t)} = 1 + \sum_{r=1}^{p-i+1} h_{t-r},$$

so its largest entry is  $h_t$ . Direct substitution proves this identity, including  $p = 1$ .

### 3 Upper bound

At the start of stage  $k$ , original rows with labels  $i \geq k + p$  are unchanged by earlier stages: their first  $k - 1$  original entries are zero. They have zero earlier multipliers and cannot be selected as nonzero earlier pivots. Thus there are at most  $p$  old surviving rows plus fresh original row  $k + p$ ; later rows are untouched. Near the bottom, missing fresh rows can be padded by zeros.

Fix target column  $j$  and set  $L = p + q$ . For  $j \geq L + 1$ , all target entries in the front are zero until stage  $k_0 = j - L$ , when original row  $j - q$  arrives. Exactly  $L$  updates occur from  $k_0$  through  $j - 1$ . Each fresh target entry has modulus at most one, so the lemma bounds every value by  $h_L$ . For  $j \leq L$ , the initial  $p$  old values are bounded by the canonical state after one imaginary update. At most  $j - 1$  actual updates precede elimination of column  $j$ , giving  $h_j \leq h_L$ . Recorded pivot-row entries and untouched entries satisfy the same bound. This covers every active entry. For  $p = 0$  the matrix is upper triangular and all elimination multipliers vanish.

### 4 Attaining construction

For  $p \geq 1$ , put  $j = p + q + 1$ ,  $n = j + p$ , and let  $L_0$  be unit lower triangular with  $-1$  on its first  $p$  subdiagonals. Use the original-row order

$$\sigma = (p + 1, 1, 2, \dots, p, p + 2, \dots, n), \quad (Pa)_i = a_{\sigma_i}, \quad \eta = 2^{-p}.$$

For  $k < j$ , define an upper-supported vector  $u^{(k)}$  by

$$u_i^{(k)} = \begin{cases} 1 & i = 1, \ k \leq p + 1, \\ 2^{i-2} & 2 \leq i \leq k \leq p + 1, \\ 1 & i = k, \ k \geq p + 2, \\ 0 & \text{otherwise,} \end{cases} \quad A_{:,k} = \eta P^T L_0 u^{(k)}.$$

Set

$$A_{i,j} = \begin{cases} 1 & p + 1 \leq i \leq n, \\ 0 & i \leq p, \end{cases} \quad A_{:,k} = e_k \quad (k > j).$$

For  $k \leq p + 1$ , the first entry of  $L_0 u^{(k)}$  is 1, entries  $2, \dots, k$  are zero, and later entries are negatives of sums of subsets of  $u^{(k)}$ . Their moduli are at most  $2^{k-1} \leq 2^p$ . After row-label assignment, support lies in original rows  $k, \dots, k + p$ . For  $k \geq p + 2$  the same support follows from  $L_0 e_k$ . Hence these columns have lower bandwidth at most  $p$  and upper bandwidth zero,

with entries bounded by one. The target column starts at  $p + 1 = j - q$  and ends at  $n = j + p$ . Later identity columns respect the band. The initial maximum is one.

In the first  $j - 1$  stages in row order  $\sigma$ , the factorization gives multipliers in  $\{0, -1\}$  and nonzero pivots. After earlier eliminations, active column  $k$  is  $L_{0,ik}U_{kk}$ ; the diagonal multiplier is one, so each chosen pivot is maximal in modulus. Thus these are admissible GEPP choices, not an extra preliminary permutation.

For the target column,  $c = PA_{:,j}$  has  $c_1 = 1$ ,  $c_i = 0$  for  $2 \leq i \leq p + 1$ , and  $c_i = 1$  for  $i \geq p + 2$ . Forward substitution gives

$$u_i = c_i + \sum_{r=1}^p u_{i-r}, \quad u_i = 0 \ (i \leq 0).$$

Induction gives  $u_1 = 1$  and  $u_i = h_{i-1}$  for  $i \geq 2$ . Therefore the active  $(j, j)$  entry at stage  $j$  equals  $h_{p+q}$ . The leading  $j \times j$  block has nonzero pivots, and later columns are identity columns; the matrix is block lower triangular with nonsingular diagonal blocks. This proves attainment.

## 5 Checks and source status

The standard-library exact verifier checks 60 attaining matrices ( $1 \leq p \leq 6$ ,  $0 \leq q \leq 9$ ), their support, pivot admissibility, growth, and nonsingularity, plus recurrence identities. These finite checks are not a formal proof of all orders.

Recovered target description: *Sharp growth for unequal lower and upper bandwidths*, provisionally IE-13 in OpenProblemsInNLA. The current repository statement could not be retrieved during reconstruction. The result is therefore submitted for review under the precise hypotheses above, not asserted to resolve a source statement whose scope has been independently verified. Repository supplied for the task:<sup>1</sup>

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<sup>1</sup><https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/linear-systems-and-elimination>.